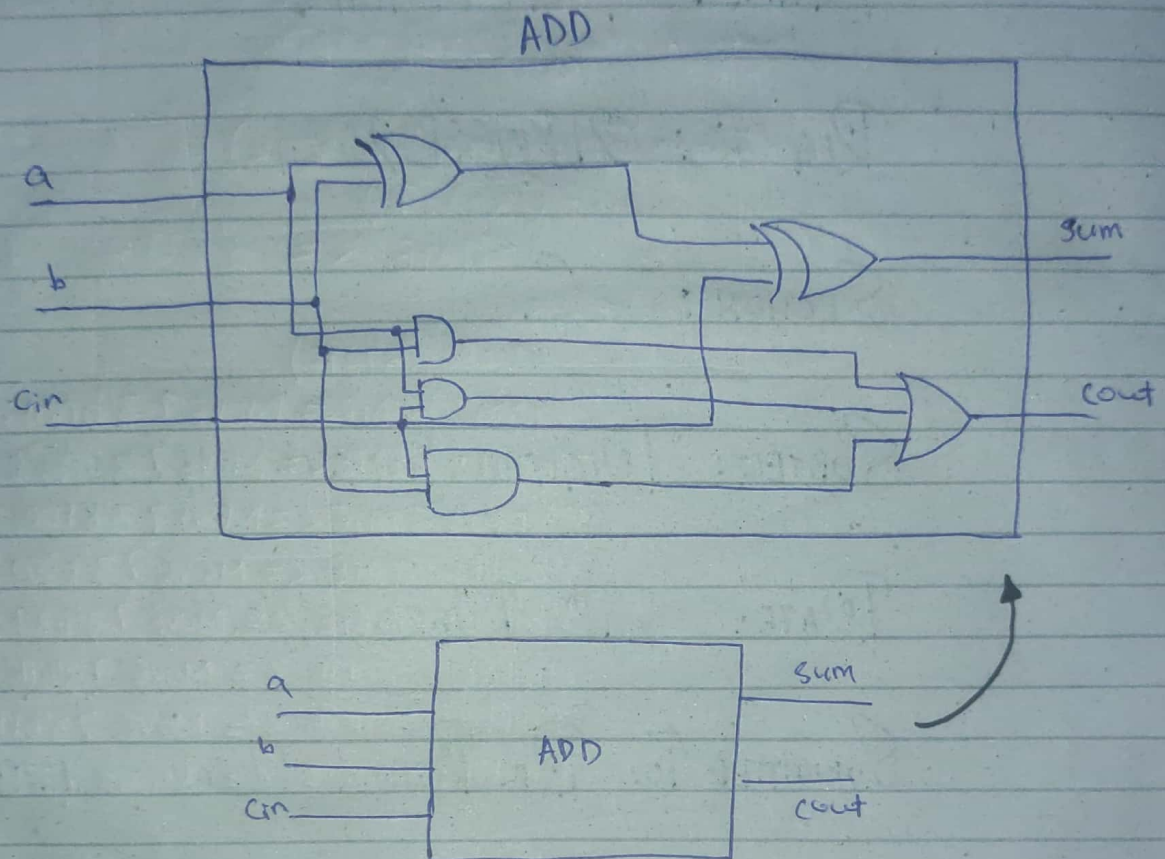
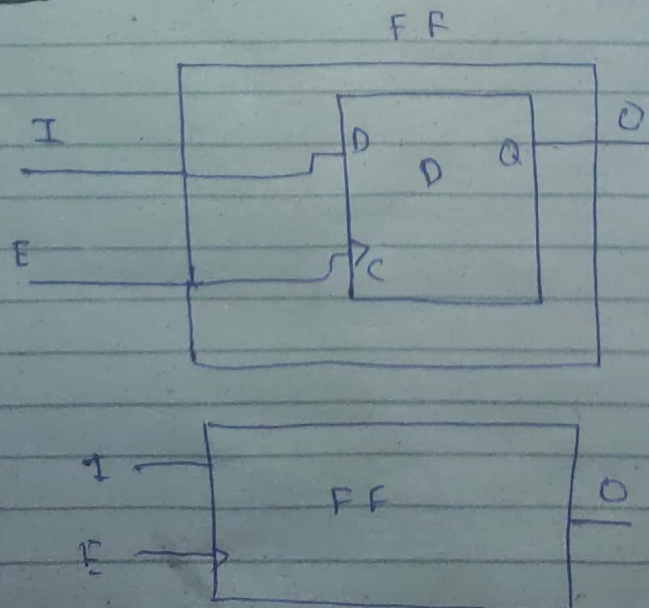


PROBLEM ④

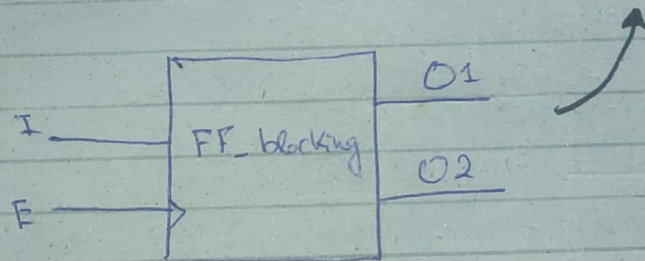
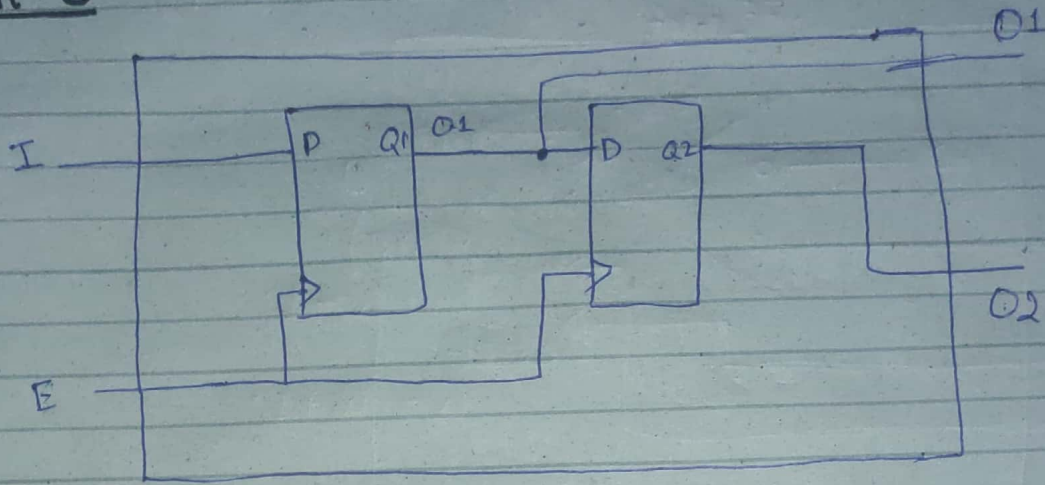
Part a



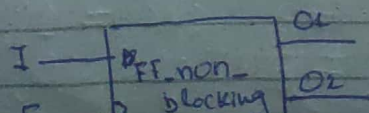
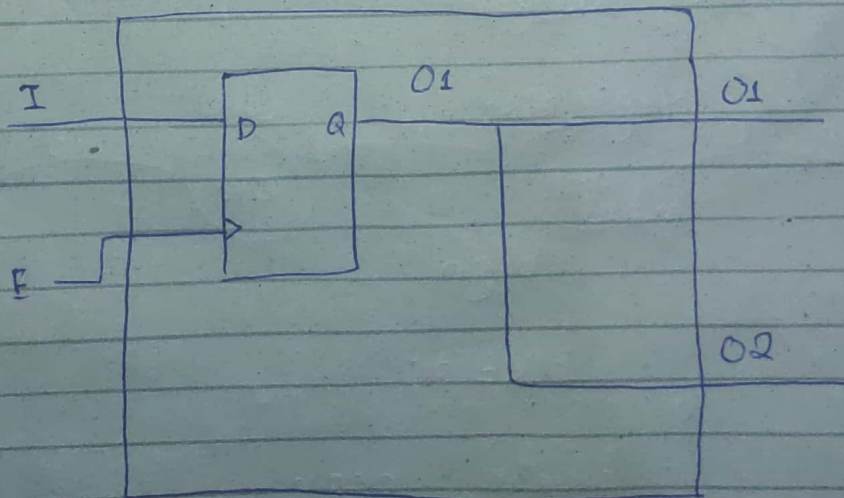
Part b



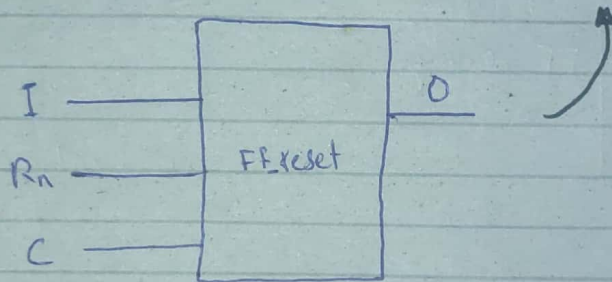
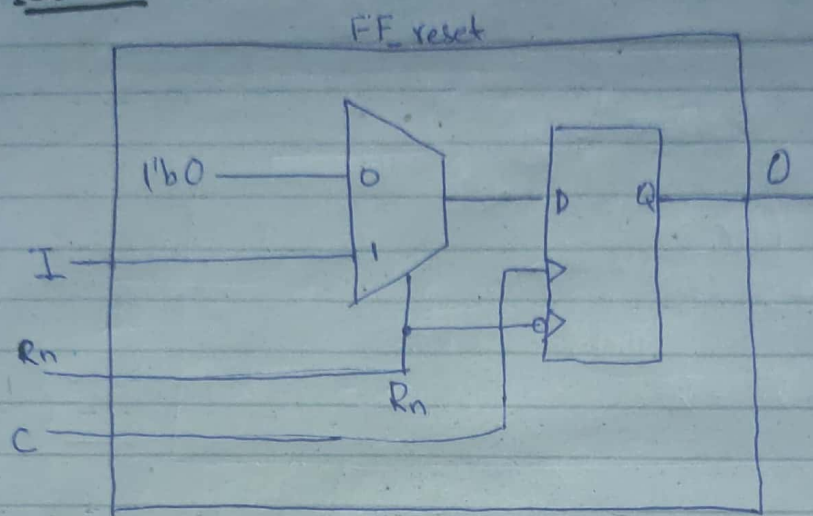
Part c.



Part d



Part e



PROBLEM 2:

```

module clk_gen
    reg clk;

    initial
        begin
            clk = 0;

            forever begin
                #10 clk = 1;
                #3  clk = 0;
            end
        end
endmodule

```


PROBLEM 3:

```
module SISO_8bit_Reg (CLK, SI, SHIFT, CLR, SO);  
  
    input CLK, SI, SHIFT, CLR;  
    output reg SO;  
  
    reg [7:0] temp;  
  
    always @ (posedge CLK)  
        if (SHIFT) begin  
            SO = temp[7];  
            temp = {temp[6:0], 1'b0};  
        end  
  
        else  
            temp[0] = SI;  
  
    always @ (CLR)  
        temp = 8'h00;  
  
endmodule
```

PROBLEM 4:

```
module Up_Down_Ctr3 (CLK, RST, LD, UP_DN,  
                    DIN, Q);  
  
    input CLK, RST, LD, UP_DN;  
    input [2:0] DIN;  
    output reg [2:0] Q;
```

always @ (posedge CLK or RST)

if (!RST)

Q = 3'd3;

else

if (UP_DN)

Q = Q - 1;

else

Q = Q + 1;

always @ (LD)

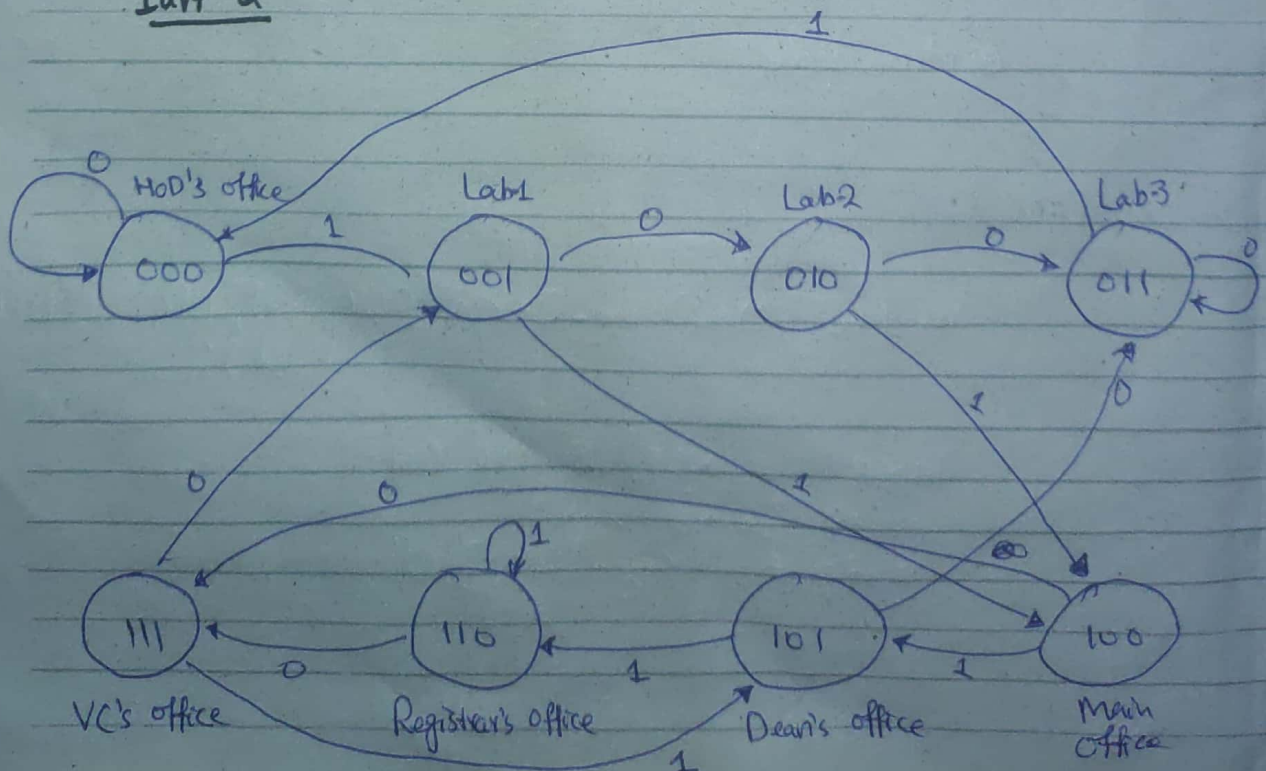
if (LD)

Q = DIN;

endmodule

PROBLEM 5

Part a



Part b

```
module PEER( MOVE, RESET, CLK, STATE);  
  input MOVE, RESET, CLK;  
  output reg [2:0] STATE;  
  
  always@ (posedge CLK)  
    if (RESET)  
      STATE <= 3'b000;  
    else  
      case (STATE)  
        3'b000: STATE <= MOVE ? 3'b001 : 3'b000;  
        3'b001: STATE <= MOVE ? 3'b010 : 3'b001;  
        3'b010: STATE <= MOVE ? 3'b100 : 3'b011;  
        3'b011: STATE <= MOVE ? 3'b000 : 3'b011;  
        3'b100: STATE <= MOVE ? 3'b101 : 3'b111;  
        3'b101: STATE <= MOVE ? 3'b110 : 3'b011;  
        3'b110: STATE <= MOVE ? 3'b110 : 3'b111;  
        3'b111: STATE <= MOVE ? 3'b101 : 3'b000;  
  
        default: STATE <= 3'b000;  
      endcase  
  
endmodule
```